

omicstra · methods (teaching companion)

TNBC-92 v3 - how every run and every number is produced

companion to `tnbc92_results_summary.md`. that doc reports *what the results are*; this doc teaches *what each run is*, *what every biological label means*, and *how each metric works* - the concept, the intuition, and the math - so someone new to spatial biology or to this project can follow the pipeline end to end and understand every number. (where the code lives is a one-line pointer at the bottom; this doc is for learning the methods, not navigating the repo.)

draft-for-voice-pass. Claude-drafted scaffolding (per memory `feedback_writing_voice`); code references verified against the scripts in `scripts/`. voice pass before publishing.

biological data H2 evaluates against (glossary)

H2 asks whether the aligned space preserves biological structure. “biological structure” is operationalized by several ground-truth labels from Wang et al. 2024, **at different biological resolutions**. the central subtlety (Cameron’s evaluation-question concern) is that some labels are **patient-level** (one value per whole tumor) and some are **niche/spot-level** (vary within a tumor) - a niche-level objective must be tested with a niche-level label, or the metric silently measures patient identity.

the ground-truth labels at a glance

label	what it is, biologically	resolution	how derived	role in H2
9 spatial archetypes (SA)	a whole-tumor profile classification, mappable from bulk-RNA molecular subtype (see SA table below)	patient-level (1 per patient)	cluster the 92 patients' global-pseudobulk ST into 9 groups	Part A proposal-literal label - diagnostic only
mc_megacluster (14 classes)	a recurring transcriptomic <i>tissue state</i> of a niche (the local cell-state program)	per-spot / niche	NMF, 14 factors, on the spot×gene matrix; each spot = its dominant factor	Part A audit-correct label
17 morphological categories	what the tissue <i>looks like</i> under H&E to a pathologist	per-spot	expert annotation (12 populated in test set: Tumor, Necrosis, High/Low-TIL stroma, Lymphoid nodule [TLS], Acellular stroma, Stroma cell, Lymphocyte, Fat tissue, Lactiferous duct, Vessels, in-situ)	Part B.1
TLS gene signature	continuous “TLS-ness” - strength of the tertiary-lymphoid-structure program (Wang’s 30-gene ST signature)	per-spot / niche (continuous)	gene-signature scoring on ST counts (tls_scores.tsv)	Part B.2

label	what it is, biologically	resolution	how derived	role in H2
Bareche TIME / MC_global / MC_tumor	patient molecular/immune subtypes from bulk RNA (TIME = tumor-immune-microenvironment class)	patient-level	Bareche bulk-RNA subtyping, propagated to all the patient's niches	Part C
mc_weights (14-d)	the <i>soft</i> version of mc_megacluster - the full NMF loading mixture per niche	per-niche (continuous)	the 14 NMF loadings, mean-pooled	supervision only (R2 positive pairs); never an H2 label or model input
patient_id	which of the 94 patients a niche came from	per-patient	metadata	the confound axis the construct-validity tests check against

the 9 spatial archetypes (Wang Fig 9) - and why they are patient-level

an archetype is built by collapsing all of a patient's spatial spots into one average (global pseudobulk) and clustering the 92 patients, so the spatial dimension is averaged away before clustering. Wang Fig 9 shows the SAs as a refinement of the 5 **bulk-RNA molecular subtypes** (each SA is mappable from bulk RNA). consequence: **every niche in patient P inherits the same archetype A_P** (verified: 14/14 held-out patients carry exactly one archetype; NMI(archetype, patient_id) = 0.89). clustering niches by archetype therefore measures “which patient is this niche from,” not “what tissue state is this niche.” -> diagnostic only; the audit-correct Part A label is mc_megacluster.

SA	characterization (Wang Fig 9)	bulk subtype origin	prognosis	actionable target
SA1		IM	-	ICBs, PARPi

SA	characterization (Wang Fig 9)	bulk subtype origin	prognosis	actionable target
	high immune, DNA repair deficiency, <i>KRAS</i> -dependent			
SA2	high proliferation, high <i>ERBB3</i> expression	IM	-	ICBs, ADCs (Trop-2, <i>ERBB3</i>)
SA3	high proliferation, high immune with <i>PD-L1</i> expression	BL	-	ICBs
SA4	high immune with TLS signature + <i>PD-L1</i> , DNA repair deficiency	BL	good	ICBs, PARPi, de-escalation
SA5	LAR-enriched, high metabolic pathways, high <i>ERBB2</i>	M / LAR	-	antiandrogens, PIK3CA/mTOR inh., ADCs (<i>ERBB2</i>)
SA6	high stromal, high proliferation, high <i>PDGF</i>	M	-	targeting PDGF signaling
SA7	high stromal, high EMT, high angiogenesis + adenosine, high proliferation	MSL	-	anti-CD73, antiangiogenic, targeting EMT
SA8	low immune, high <i>NECTIN4</i> expression	LAR	poor	ADCs (<i>NECTIN4</i>)
SA9	high stromal, low immune	LAR	-	targeting stroma

patient counts (this cohort): SA4=16, SA7=15, SA2=14, SA5=13, SA6/SA9/SA3=8 each, SA8/SA1=6 each. **SA4 is the TLS archetype** (organized immune response, good prognosis - Wang's headline, and the tie-in to H2 Part B); **SA8** (NECTIN4-high) is the poor-prognosis archetype.

“bulk subtype origin” - the 5 pre-existing TNBC molecular subtypes defined from bulk RNA-seq (no spatial info); Fig 9 is a Sankey flowing each tumor from its bulk subtype to its SA. the mapping is **not 1:1** - one bulk subtype splits across several SAs (e.g. LAR -> SA5/SA8/SA9), and that splitting is the added value of the spatial archetypes. the column shows the dominant flow, not a strict assignment.

abbr	bulk molecular subtype
IM	Immunomodulatory
BL	Basal-Like
M	Mesenchymal
MSL	Mesenchymal Stem-Like
LAR	Luminal Androgen Receptor

“prognosis” - from Wang Fig 8g (iBCFS multivariate survival, combined cohort, adjusted for age/size/nodal status). only two SAs reached significance: **SA4 good** (HR 0.60, FDR 0.047), **SA8 poor** (HR 1.8, FDR 0.039); the other seven are “-” (no significant outcome association).

“actionable target” - Wang Fig 9 “therapeutic perspectives”: the precision-medicine strategy each archetype's molecular features point to.

abbr	therapy	rationale
ICBs	immune checkpoint blockade (anti-PD-1/ PD-L1)	high-immune / PD-L1+ archetypes (SA1-4)
PARPi	PARP inhibitors	DNA-repair-deficient archetypes (SA1, SA4)
ADCs	antibody-drug conjugates	target a surface protein: Trop-2 / <i>ERBB3</i> (SA2), <i>ERBB2</i> (SA5), <i>NECTIN4</i> (SA8)
antiandrogens, PIK3CA/mTOR inh.	hormone + pathway inhibitors	LAR / AR+ archetype (SA5) SA7

abbr	therapy	rationale
anti-CD73, antiangiogenic, EMT- targeting	stromal / mesenchymal therapies	
PDGF / stroma targeting	stromal therapies	SA6, SA9

these are all **whole-tumor** characterizations (origin, outcome, therapy)
- none is a spatial/regional property, which is again why the archetype
label is patient-level.

TLS - tertiary lymphoid structure (the one genuine FTU)

an organized ectopic lymphoid aggregate inside the tumor: a B-cell follicle + adjacent T-cell zone + high endothelial venules, performing local antigen presentation and antibody maturation - an immune “mini lymph node” the tumor grows. it is a **functional tissue unit (FTU)**: recurring, sharply bounded, defined function. clinically, TLS density associates with immunotherapy response in TNBC (Wang derived a 30-gene TLS signature for exactly this). the dataset captures it two ways - the categorical “**Lymphoid nodule**” pathologist annotation (Part B.1) and the continuous **TLS gene signature** (Part B.2) - and it is the one compartment where H2 Part B’s directional prediction held.

mc_megacluster - the audit-correct niche-level biology

Wang’s per-spot 14-class NMF. NMF decomposes the spot × gene matrix into 14 non-negative factors; each factor is a recurring cell-state program, and each spot is assigned the factor it loads on most. unlike archetype, these states **recur within and across patients** - a single tumor section contains several megaclusters - so clustering niches by mc_megacluster is a real niche-level biology test. (exact factor identities: Wang’s mc characterization, data/inputs/clustering/clustPrototypes/.)

0 · the pipeline in one picture

raw inputs	per-niche features
shared 512-d space	evaluation
-----	-----
-----	-----
H&E WSI --Virchow2-->	virchow2_niche (1280-d) --\

```

(frozen FM)                virchow2_cell_tokens (7x1280)
\                            H1: retrieval (AUC, R@K, CKA)
                             >--
align.py --> z_he, z_st -->  H2: clustering (ARI, probe, cosine)
ST counts --Novae GNN-> novae_niche (64-d) -----/ (MLP or
cross-attn)                H3: pathway CCA (z_A, BH-FDR)
(frozen FM)                gpath2vec_niche (512-d) ---/
InfoNCE / SupCon / Barlow
                             mc_weights_niche (14-d, supervision only)
                             tls (1-d), mc_megacluster (label),
compartiment (label)

```

stage 1 - niche-join (scripts/build_niche_join.py): for each subarray, build one row per niche (center spot + 6 spatial neighbors). mean-pool Novae over the 7 spots then z-score per subarray; keep Virchow2 niche-mean AND the 7 raw tile tokens; attach the v3 gpath2vec embedding; attach labels (mc_megacluster, compartiment, tls, mc_weights). output: data/embeddings/niches_v3/{subarray}.parquet. niches with no v3 gpath2vec coverage are dropped (intersection split) -> 208,786 niches.

stage 2 - alignment (scripts/align.py for contrastive R-runs, scripts/align_classical.py for B1-B3, scripts/align_b4.py for B4): project both modalities into a shared 512-d space. patient-stratified 85/15 split (seed 42) -> train on 80 patients, test on 14. output per run: runs/tnbc-92_v3/{run}/embeddings_test.parquet with per-niche z_he, z_st (and attention_weights for R4).

stage 3 - evaluation (scripts/eval.py + the per-hypothesis scripts): compute H1/H2/H3 metrics on the 35,594 held-out test niches.

the input features defined (what feeds the towers)

a **niche** = a center spot + its 6 spatial neighbors = 7 spots (~1200 cells). every run sees the same four per-niche features; they differ only in which they consume and how they fuse them.

feature	dim	what it is	what it captures
virchow2_niche	1280	the 7 spots' Virchow2 foundation-model embeddings mean-pooled into one vector	the <i>average</i> H&E morphology of the niche. used by every run except R4.
virchow2_cell_tokens			

feature	dim	what it is	what it captures
	7 x 1280	the same 7 per-spot Virchow2 embeddings kept separate (not pooled)	within-niche morphological detail - lets R4's cross-attention weight each spot instead of averaging. R4 only.
novae_niche	64	the Novae GNN (GAT) latent, mean-pooled over the 7 spots then z-scored per subarray	spatially-aware ST <i>expression</i> : what the niche expresses given its spatial transcriptomic neighborhood. <i>which Reactome biological processes are on</i> in the niche. built by Fisher-enriching each niche's expression against Reactome, then
gpath2vec_niche	512	a metapath2vec embedding of the niche's pathway-activation profile (v3 build fisher_madmean_low_dim512_e5_s1234, sha-locked)	metapath2vec random walks over the niche<->pathway graph - so niches with similar active-pathway

feature	dim	what it is	what it captures
			programs land close together. complements novae (program-level, not raw-expression-level).

virchow2_niche vs the 7 tile tokens is the single H&E difference across runs: pooling (everyone) throws away which of the 7 spots drove the signal; R4 keeps them addressable so attention can recover it. that trade is R4's whole bet, and it wins H1 + H3.

the two + in the run table are different operations. when a run's input reads `Virchow2 niche + novae+gpath2vec`:

- the **inner + (novae + gpath2vec)** is a true **concatenation** -> one 576-d ST feature vector (64 + 512). (`align.py:146, x_st = np.concatenate(...)`; written $\oplus$ in the run tables below.)
- the **outer + (Virchow2 ... + novae+gpath2vec)** is **not** a concatenation. the H&E vector and the ST vector are the two *sides* of the contrastive pair: each passes through its **own MLP tower** into the shared 512-d space, where InfoNCE/SupCon/Barlow pulls matched (H&E, ST) niches together. the two modalities are never stacked into a single vector.

1 · run definitions (all 10)

every run maps the same two inputs (H&E side, ST side) into a shared 512-d space; they differ in **loss**, **fusion architecture**, and **ST input composition**.

the shared building block: `MLPBlock (align.py)`

what an MLP is. an **MLP** (multi-layer perceptron) is the simplest learnable neural network: a stack of `Linear` layers (each a learned weight matrix) with nonlinearities (`ReLU`) between them. an MLP turns one vector into another vector by applying `x -> Linear -> nonlinearity -> Linear -> ...`; the weights are learned from data. an

MLP tower is one such stack used as a column that processes one modality end-to-end. in “late fusion” each modality gets its **own** tower with independent weights (no sharing): the H&E vector goes through MLPBlock_HE, the ST vector through MLPBlock_ST, each produces a 512-d output, and the contrastive loss aligns them in that shared space.

the specific MLPBlock used here:

```
LayerNorm(in) -> Linear(in, 512) -> ReLU -> BatchNorm1d(512) ->
Dropout(0.3) -> Linear(512, 512) -> L2-normalize
```

- two Linear layers = two learned weight matrices.
- LayerNorm / BatchNorm1d stabilize training by normalizing activations.
- Dropout(0.3) randomly zeros 30% of activations during training (regularization).
- L2-normalize at the end puts every output on the unit sphere, so downstream “similarity” = cosine = dot product.

every learned run uses this block (one per modality for late fusion; R4 replaces the H&E tower with CrossAttnFusion).

contrastive runs (trained with SGD, align.py)

run	loss	fusion	ST input	plain-English what it is
R1	InfoNCE	late (2 independent MLPBlocks)	novae(64) $\oplus$ gpath2vec(512) = 576-d	the standard CLIP-style baseline. H&E niche-mean -> MLP; ST -> MLP; pull matched (H&E, ST) niche pairs together, push cross-patient mismatches apart.
R2	SupCon	late	novae $\oplus$ gpath2vec	like R1 but the “correct match” target is <i>soft</i> : niches with similar

run	loss	fusion	ST input	plain-English what it is
				mc_weights (Wang's 14-d cell-state mixture) are treated as partial positives, not just the exact diagonal.
R3	Barlow Twins	late	novae $\oplus$ gpath2vec	no negatives at all. drives the H&E $\times$ ST cross- correlation matrix toward the identity (matched dimensions correlated, off- diagonals decorrelated).
R4	InfoNCE	cross- attention	novae $\oplus$ gpath2vec	the only non- late run. the ST embedding <i>queries</i> the 7 raw H&E tile tokens (center + 6 neighbors) via attention, producing a weighted-sum H&E representation conditioned on ST.
R5	AnInfoNCE	late		captures which tiles matter instead of mean- pooling them.

run	loss	fusion	ST input	plain-English what it is
			novae $\oplus$ gpath2vec	R1 + a learnable per-dimension temperature on the bilinear similarity (diagonal Mahalanobis). lets the loss weight some axes of the shared space harder than others. init = 0 -> identical to InfoNCE at step 0. the ablation: drop gpath2vec from the ST input. tests whether the H&E pathway signal needs a pathway anchor on the ST side.
R6	InfoNCE	late	novae only (64-d)	

baselines

run	what it is	trained?	why it exists
B1	Canonical Correlation Analysis (CCA) - closed-form. find linear projections of H&E and ST	closed-form (no SGD)	the proposal's named "CCA projection" baseline. does classical linear correlation suffice?

run	what it is	trained?	why it exists
	with maximum cross-modal correlation in whitened space.		
B2	Orthogonal Procrustes - PCA each modality to 512-d, then solve for the orthogonal rotation minimizing paired Frobenius distance.	closed-form	is a rigid rotation enough, without whitening?
B3	Unaligned PCA - independent per-modality PCA to 512-d, L2-norm, no cross-modal step .	closed-form	the proposal's "unaligned concatenation" sanity floor. any real method must beat this; if B3 scores high, the test is broken.
B4	Random-init MLP - R1's exact architecture, Xavier init, zero training steps .	not trained	the strict control: separates "is R1's gain from the <i>loss</i> ?" from "is it from the <i>architecture shape</i> ?" if B4 $\approx$ chance, the loss is doing the work.

the circular-dependency guard (why mc_weights is supervision-only)

mc_weights_niche (Wang's 14-d cell-state mixture) defines the *positive pairs* for R2's SupCon loss. it is therefore **never** put in the ST input vector - if it were both the input and the target, the model would predict its own input. align.py hard-asserts mc_weights_niche $\notin$ st_features (program.md constraint 3).

2 · how each metric works (concept, intuition, math)

each metric below is explained as a *method* - what it measures, the intuition, and the computation - not as a code location. (the functions that implement them are listed once at the end of this section for anyone who wants to read the source.)

H1 retrieval metrics

H1 asks: can an H&E niche find its matched ST niche among all the others, in patients the model never trained on? all H1 numbers operate on the L2-normalized z_{he} and z_{st} matrices ($N \times 512$) - every embedding is a unit vector, so similarity = cosine = dot product.

```
sim = z_he @ z_st.T          # (N, N) cosine-similarity matrix;
entry (i,j) = cos(H&E_i, ST_j)
ranks = (-sim).argsort(axis=1).argsort(axis=1)[diag]  # 0-indexed
rank of the matched ST for each H&E query
```

- **R@1 / R@5 / R@10** = fraction of queries whose matched ST falls in the top 1 / 5 / 10. $(ranks < K).mean()$.
- **MRR** (mean reciprocal rank) = $mean(1 / (ranks + 1))$.
- **median rank** = $median(ranks + 1)$. random baseline $\approx N/2 = 17,797$.
- **alignment gap** = $mean(diagonal\ cosines) - mean(off-diagonal\ cosines)$. positive = matched pairs systematically closer.
- **AUC** - the headline H1 number. computed with **scikit-learn** roc_auc_score, NOT a nearest-neighbor method:

```
matched      = diag(sim)          # cosine of each
true (H&E, ST) pair
```

```

mismatched = sim[off-diagonal], sampled to 10x len(matched)
for speed
labels      = [1]*len(matched) + [0]*len(mismatched)
scores      = concat(matched, mismatched)
auc         = roc_auc_score(labels, scores)

```

interpretation: probability that a random matched pair has higher cosine than a random mismatched pair. 0.5 = chance, 1.0 = perfect. it's a ranking metric over cosines - no clustering, no kNN.

H1 representation structure (CKA)

CKA (Centered Kernel Alignment, linear kernel) measures whether the two modality embeddings have the same *pairwise-similarity geometry*, even if individual pairs aren't matched:

```

Xc = X - X.mean(0); Yc = Y - Y.mean(0)           # center
num = ||Xc.T @ Yc||_F^2                          # HSIC(X, Y)
den = ||Xc.T @ Xc||_F * ||Yc.T @ Yc||_F
CKA = num / den                                   # 0 = unrelated
geometry, 1 = identical

```

cka_before is computed on the raw inputs (virchow2_niche vs raw ST features); cka_after on the aligned z_he vs z_st.

H2 Part A clustering (ARI, silhouette)

ARI (Adjusted Rand Index) on KMeans clusters vs a biology label:

```

km    = KMeans(n_clusters=14, n_init=10, random_state=42)  #
sklearn
preds = km.fit_predict(z_he)
ari    = adjusted_rand_score(y_true_label, preds)          #
sklearn; chance-corrected, 0=random 1=perfect

```

the label y is mc_megacluster (audit-correct) or archetype (proposal-literal, diagnostic only). **silhouette** = silhouette_score on a 5,000-niche subsample (cosine geometry). this is a clustering metric - KMeans assumes globular clusters, which is why R4's tight non-globular manifold scores low here even though its biology is linearly recoverable (see linear probe).

H2 Part A diagnostics (kNN purity, linear probe, rank-matched KMeans)

three metric-independent cross-checks on the same `mc_megacluster` label:

- **kNN purity** (`knn_purity()`): L2-normalize `z_he`, fit **sklearn** `NearestNeighbors` (euclidean on unit vectors = cosine), for each niche check what fraction of its `k` nearest neighbors share its `mc_megacluster`. reported at `k = 5/10/25`. chance = $1/14 = 0.071$. caveat: dominated by within-patient niche similarity - not patient-stratified, so it's patient-confounded.
- **linear probe** (`linear_probe()`): the metric-independent biology test. split the 14 test patients into 11 probe-train / 3 probe-test (patient-level, so no patient leaks across the split). `StandardScaler` on train, `LogisticRegression(max_iter=300, lbfgs, C=1.0)` 14-class, score accuracy + macro-F1 on the held-out 3 patients. chance = 0.071. **this is what shows R4 encodes mc biology at parity** even though its KMeans ARI is low - LogReg needs only linear separability, not globular clusters.
- **rank-matched KMeans**: project `z_he` to its top-3 PCs (SVD), recompute ARI. controls for R4's low effective dimension - if a wider run at rank-3 still beats R4 full, R4's deficit is encoding-shape, not just dimensionality.

H2 Part B.1 compartment contrasts (Welch z-test)

`per(run, compartment), eval.py` aggregates matched-pair cosine `cos(z_he_i, z_st_i)` to `{n, mean, std}`. the **Welch z-test** (unpooled variance) on the 6 named contrasts:

```
z = (mean_distinct - mean_ambiguous) / sqrt(std_distinct^2 / n_distinct + std_ambiguous^2 / n_ambiguous)
```

`pass = z ≥ +1.96` (distinct > ambiguous, the proposal direction); inverted = `z ≤ -1.96`. Welch (not pooled-variance) because compartments have very different cell counts and variance.

H2 Part B.2 TLS gene-signature (univariate CCA)

direct test against the continuous per-niche TLS score (`tls_scores.tsv`, mean-pooled to niche). **univariate CCA** between `z_he` (512-d) and the 1-d TLS score:


```

w          = lstsq(Z_centered, y_centered)      # normal-equations
solution for the canonical direction
w          = w / ||w||                          # unit-norm
proj       = Z_centered @ w
r          = pearsonr(proj, y_centered)         # the canonical
correlation (1-d target -> reduces to pearson r)

```

fit on 11 train patients, evaluate the *frozen* w on 3 held-out patients.

permutation null: shuffle the held-out TLS scores 500×, recompute r each time; $z_{\text{TLS}} = (r_{\text{test}} - \text{null_mean}) / \text{null_std}$. significance by **BH-FDR** across the run grid.

H3 pathway CCA (cross-patient transfer)

same univariate-CCA machinery as B.2, but the 1-d target is a *pathway* signal (AUCell score, or gpath2vec niche-cosine-to-pathway-subtree).

two operationalizations: - **option B (fit-on-all):** fit and evaluate on all test niches. proposal-literal but universally positive (doesn't discriminate). - **option A (cross-patient):** fit on 11 train patients, evaluate frozen w on 3 held-out patients. **this is the discriminating test.** z_A = held-out r in permutation-null std units.

permutation null (eval_h3_pathway_cca_perms.py): for speed, the canonical correlation = $\sqrt{R^2}$ of the OLS regression, and $R^2 = ||Q^T y||^2 / ||y||^2$ where $Z = QR$ (skinny QR). each permutation is one $\text{matmul } Q^T y_{\text{shuffled}}$, amortizing Q across the 5 pathways. 500-1000 perms per cell. **BH-FDR** applied within each declared test family (e.g., the testable z_{he} pathways).

BH-FDR (multiple-testing correction)

Benjamini-Hochberg: sort p-values ascending, multiply the k-th by n/k , enforce monotonicity from the top, clip to [0,1]. controls the false-discovery rate across a family of tests so that “X/N significant” means something across many simultaneous comparisons (e.g. the 120-cell H3 grid, or the per-run TLS grid).

implementations (for the code-curious)

all metrics use standard scikit-learn (roc_auc_score, KMeans, adjusted_rand_score, LogisticRegression, NearestNeighbors) and scipy (pearsonr) - auditable building blocks. **the full calculation-provenance map - which script + function computes each metric,**

and which artifact each result lands in - is its own doc: [tnbc92_provenance.md](#). (kept separate so this doc stays about the methods, not the repo layout.)

3 · the two pathway representations (biology, not code)

the project uses two different per-niche pathway encodings, and they capture different biology - this matters for reading H3.

	gpath2vec (ST input feature)	AUCell (H3 evaluation readout)
pipeline	TF-gene filter -> MAD/mean variable-gene selection per niche -> Fisher enrichment vs Reactome -> metapath2vec random walks on the (niche, pathway) graph -> 512-d	rank all genes in a niche by expression -> for each Reactome gene-set, area under the recovery curve over the top ~5% (decoupler dc.mt.auccell)
biology captured	regulatory-program neighborhood structure - which TF-driven pathways co-fire, embedded so niches with similar regulatory state sit close	direct pathway activity - how strongly a specific gene-set's genes are over-expressed in a niche
gene selection	TF-restricted + most-variable (MAD/mean)	none; all genes ranked, gene-set decides what counts
emphasizes	discriminative regulatory state	abundant pathway-gene expression (rank-based)
per-niche output	512-d dense embedding (geometry meaningful, dims not	k-d score vector, one interpretable activity score per named pathway

gpath2vec (ST input feature)	AUCCell (H3 evaluation readout)
individually interpretable)	

gpath2vec is what the H&E side learns to align *to* (it's on the ST input). AUCCell is one of the *readouts* H3 tests the aligned space against. The H3 hypothesis is only meaningful because these are different views - the alignment is informative only if organizing one (gpath2vec on ST) produces a space that predicts the other (AUCCell / TLS signature) in held-out patients.

4 · what each hypothesis tests

hypothesis	the question (plain English)	metric
H1	can H&E find its matched ST niche in patients the model never saw?	AUC, R@K, MRR, median rank, alignment gap, CKA
H2 Part A	does the aligned space group niches by transcriptomic tissue state?	KMeans ARI + silhouette (vs mc_megacluster); kNN purity, linear-probe accuracy, rank-matched ARI
H2 Part B.1	do morphologically distinct compartments cohere more than ambiguous stroma?	Welch z on 6 named contrasts
H2 Part B.2	does the H&E view predict the continuous TLS gene signature cross-patient?	univariate CCA z_TLS + permutation null + BH-FDR
H2 Part C	does the space encode biology more than patient identity?	bio-z / patient-z ratio (permutation z-tests)
H3		

hypothesis	the question (plain English)	metric
	do named Reactome pathways correlate with shared-latent directions, and transfer across held-out patients?	univariate CCA z_A (cross-patient) + permutation null + BH-FDR

5 · ground-truth labels (the four proposal-named sources)

proposal ground truth	dataset asset	resolution	used in
9 spatial archetypes	Wang Spatial archetypes_defined_on_ST_global_pseudobulk	patient-level (1 per patient)	H2 Part A literal - diagnostic only (patient confound)
TNBC molecular subtypes per spot	mc_labels.megacluster (14-class NMF)	per-spot / niche	H2 Part A audit-corr (KMeans A linear pro
17 morphological categories	morphology_labels.tsv (pathologist)	per-spot	H2 Part B. (compart cosine)
TLS gene signatures scored spatially	tls_scores.tsv (continuous)	per-spot / niche	H2 Part B. (TLS signature CCA)

the construct-validity correction in H2 Part A (archetype -> mc_megacluster) is a switch between two *proposal-named* labels at different resolutions - proposal-compliant, not a deviation.

pointers

results companion: [tnbc92_results_summary.md](#). construct-validity
audit: projects/tnbc-92/evaluation_question_audit.md. agent design:
docs/mcp_agent_design.md. code: scripts/
{build_niche_join,align,align_classical,align_b4,eval}.py +
scripts/_scratch/eval_h2_*.py + scripts/eval_h3_pathway_cca*.py.